NFL BLIND FLAG FOOTBALL

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MOTIVATIONS

Approximately **1 million** people in America experience blindness or low-vision conditions.

Only **20%** have tried adaptive recreational sports.

Our team is developing the world's first rechargeable football with variable haptic and auditory feedback as part of a mission to increase accessibility and inclusivity in sports.

This technology enables people with varying levels of blindness to play an adapted version of flag football by signaling the ball's location and state.

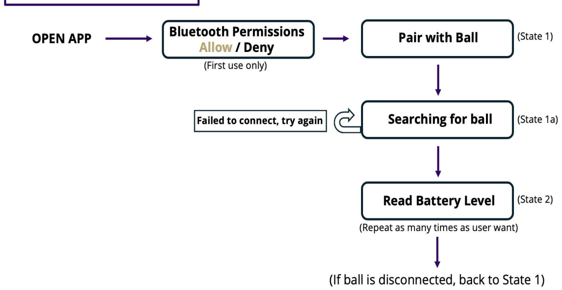
REQUIREMENTS

BEEPING
- Airborne state mode-switching
- Multi-module for audibility
- Hearing safety (SPL ≤85 dBA)

HAPTICS
- Proximity-activated vibration
- Discernible feedback (150 – 300 Hz)

CHARGING
- Battery safety
- Accessible on/off switch
- Battery status on mobile device

BALL CHARGING MOBILE APP



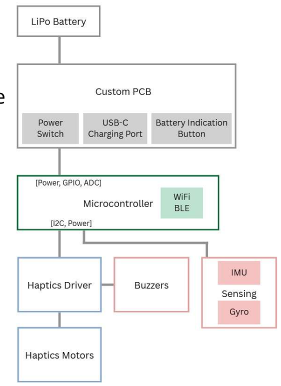
DESIGN & IMPLEMENTATION

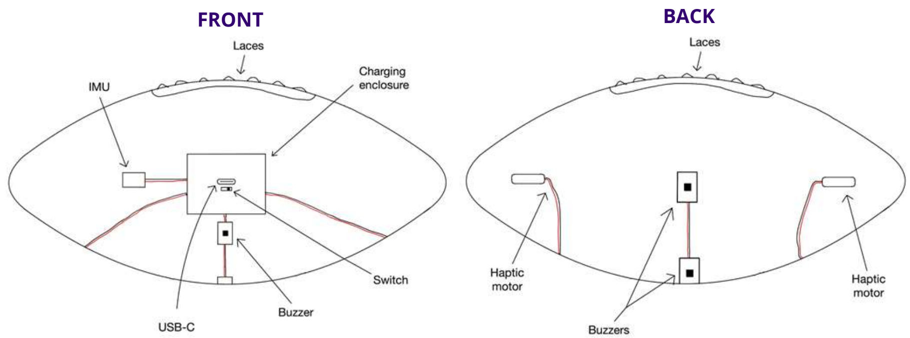
Integration:
The system components are integrated with the Raspberry Pi Pico 2W and a custom PCB board, enabling programming of complex features

Airborne Mode-Switching:
- Emits low-frequency drone when grounded and high-frequency pulse beeping when airborne with the Modulino Buzzer
- Detects airborne state using calibrated transition points for acceleration and angular velocity magnitudes measured with Modulino Movement

Proximity Detection:
- Uses blue-tooth signal attenuation (BLE RSSI) to detect distance between ball and field element (end zone, out of bounds, etc.)
- Haptic vibrations begin within 10 meters and frequency increases with proximity

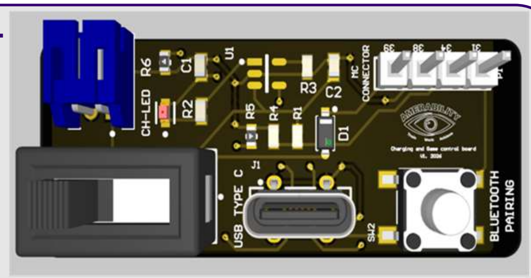
Battery Indication:
- The battery level of the ball is sampled with ADC measurements and displayed on a paired accessible app interface





CUSTOM CHARGING CIRCUIT

POWER SWITCH
USB-C CONNECTOR
BLUTOOTH PAIRING BUTTON
MC CONNECTOR
LED INDICATOR
JST CONNECTOR



RESULTS

- Early prototyping involved constructing circuitry to produce sound with the piezo element buzzer and programming haptic motors with different motors
- Conducted component trade-off analyses for durability, reliability, and feedback perceptibility
- Incorporated user feedback on integral features, haptic & sound thresholds, and switch accessibility
- Preliminary airborne mode-switching faced issues with false-positive and false-negative triggers → resolved with time-based debouncing and calibrating thresholds for the 6-axis IMU
- Performed trial testing to quantify accuracy of airborne mode-switching and proximity-activated vibration
- Distance sensing options included Wi-Fi and Bluetooth RSSI and ToF (Time of Flight) and after testing, Bluetooth RSSI was the most precise and accurate option
- By averaging RSSI samples, the distance sensing has an accuracy of $\pm$ 3 meters

CONCLUSION & FUTURE WORK

Despite football being the biggest American sport, accessibility in football is an underexplored field. Our work aims to change the landscape of accessible sports as the first ball to include haptic feedback. We designed our system to accommodate future developments so that this multi-year project will result in the production of commercial footballs for league and at-home use.

Next Steps
- UWB module and PCB integration
- Durability
- Manufacture pocket inside the ball
- App development
- Field elements (end zone, boundaries)
- Player elements (wearables)
- Work with NFL

ELECTRICAL & COMPUTER ENGINEERING
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